
CLASSIFIED // OPERATIVE DOCUMENTATION

GROUP DESIGN DOCUMENT

HEAT SIGNATURE

— CLONE PROJECT —

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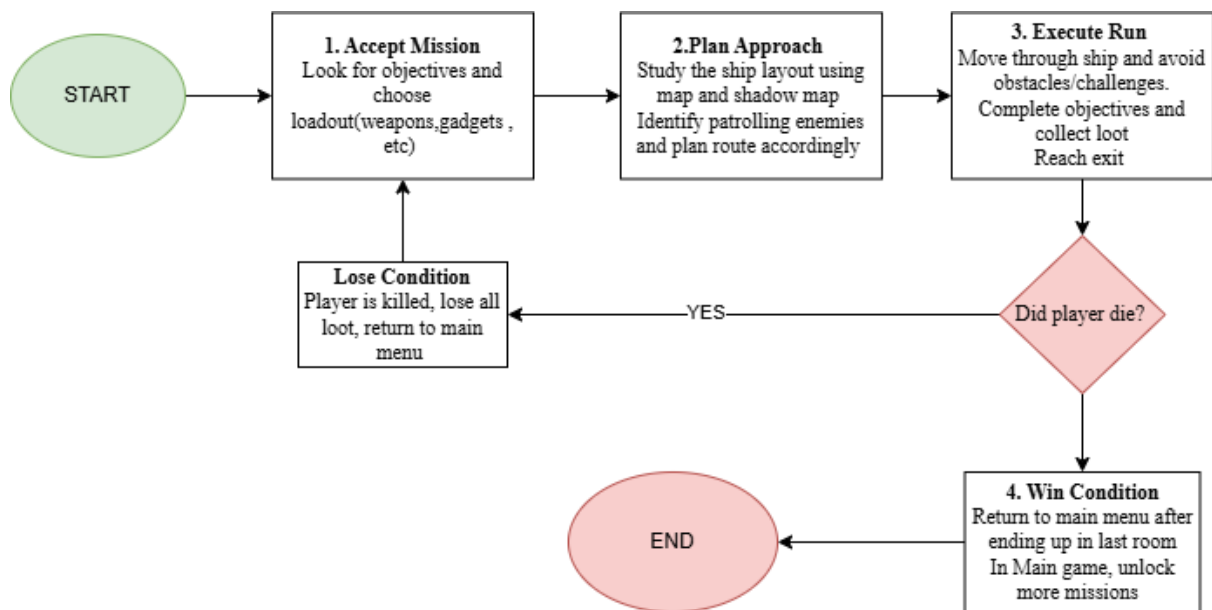
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Heat Signature is an action, 2D top down, stealth, rogue-like where you take on missions to infiltrate ships to complete different objectives on that ship. The player's final goal is to get enough influence to liberate the entire galaxy of the four different factions. Our group will be trying to clone the main mechanics of this game using Unity Engine: 6000.5.7f1.

Gameplay Loop



(Figure 1; flowchart of heat signature game loop)

Action Genre

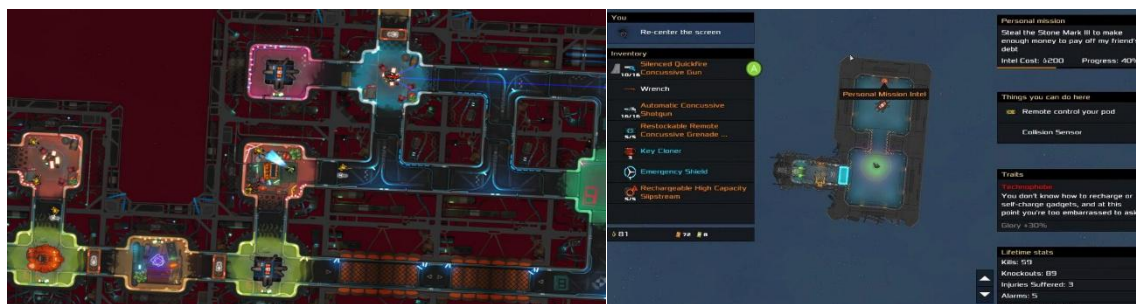
The action genre requires the player to rely on their reaction time, hand-eye coordination, and ability to respond quickly to obstacles. Players are generally expected to battle through levels while overcoming enemies and other challenges.

Heat Signature is therefore classified as an action game because the player is constantly expected to react to the obstacles and threats they encounter on the ship. This can involve reacting to enemies shooting or attacking them, using the pause mechanic to quickly assess the situation, or using gadgets to escape or change the situation to benefit them. The player can also react to incoming attacks by changing their movement or positioning to dodge them. These mechanics make quick decision-making and reacting to changing situations an important part of the gameplay.

Stealth Genre

Stealth games require the player to hide and avoid enemies through the use of different mechanics. Players are encouraged to avoid the enemies as they are typically stronger than the player character.

Heat Signature is therefore classified as a stealth game because the player gets taken out in one hit from enemies and vice versa. This creates situations where the player needs to hide or be strategic about taking out groups of enemies.



(Figure 2: An image showing a map and inventory of Heat Signature)

Interrogation

Design Question

Will replacing Heat Signature's pause-to-plan mechanic, which makes the player invisible to enemies and stops them from interacting with the map, with a ShadowMap mechanic encourage a more active style of gameplay, where players must constantly move, react to threats, and make decisions on the spot?

Hypothesis

Heat Signature's pause menu allows players to stop time and carefully plan their next move using a static map. We are questioning whether this pause-to-plan design is essential to the game's core tension and gameplay, or whether it can be replaced with a more active alternative.

Our hypothesis is that removing the pause system and replacing it with a shadow map that the player can move to and view, but with the restriction that they cannot

interact with anything while doing so , will shift the player's experience from deliberate tactical planning to reactive, in-the-moment decision-making. Rather than pausing to think through the next step, the player is forced to keep moving and react under pressure, changing how strategy and risk are experienced during gameplay.

Design Goals

Create the game's base mechanics (movement, detection, combat, the shadow map) as modular units that work together rather than isolated features so that each mechanic can be tested, iterated, and visually tracked, thus giving a clear view of development progress and how systems interact.

To test this hypothesis, we will see how players adapt or change their playstyle while interacting with the clone game when they realise that they can no longer pause to plan . We can do this to evaluate whether the change produces the intended shift from tactical planning to reactive decision-making that we are testing in hypothesis.

Cloned Game/ System

Actions and Challenges in the Original Game

Heat Signature is one of those games that gives the player more actions than typical 2D Shooter games. However, every action available to the player exists to overcome a specific obstacle, while each challenge encourages the player to make effective use of the available mechanics. This means that since there are many actions, there are also many challenges in this game.

Required Actions

- Moving with WASD.
- The player is required to move through the map to complete the mission.

Allowed Actions

- Using items with left or right mouse click.
 - The player can use any item as long as the item is off cooldown, has charges to use and the target of the item exists.
- Pausing the game.

- The player can pause the game to reorder items in their inventory for their current situation.
- Interacting with the world.
 - The player can interact with the world by pressing e. This includes stealing keycards, opening doors, picking up items and enemies.
 - The player can break windows on the map.
- Interacting with the camera.
 - The player can move the camera over the map by edge panning and holding LShift.
 - The player can zoom the camera in and out with the scroll wheel.

Actions and challenges in clone game

Required Actions

- Moving with WASD.
- The player is required to move through the map to complete the mission

Allowed Actions

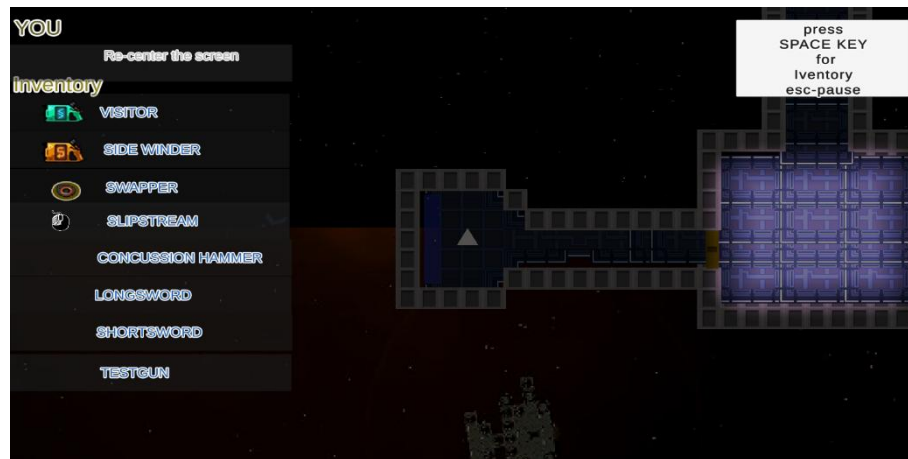
- Using items with left or right click.
 - The player can use any item as long as the item is off cooldown, has charges to use and the target of the item exists.
- Toggling the ShadowMap
 - The player can toggle a ShadowMap through pressing space in order to reorder their inventory items.
 - While in the ShadowMap the player cannot interact with the world, but the player can use items which will take them out of the ShadowMap.
- Interacting with the world
 - The player can pick up items, fight enemies and open doors.

What are we cloning from Heat Signature

The general of what we are cloning from Heat Signature is a one large map where the player will have to get past multiple enemies using limited items.

Maps

The map that we cloned is based on one of the beginner level maps of Heat Signature and will be one large map where the player will need to get past multiple patrolling enemies to complete the map. Doors will also be able interactable items.



(figure 3; map of the game and the shadow map)

Shadow Map

The shadowmap is a mechanic that allows the player to open their inventory while also becoming invisible to enemies and stopping the player from interacting with the normal map. The player can only be in the shadowmap for a limited time which forces the player to quickly change which items they are using before they are returned to the actual map.

The shadowmap replaces the pause system in the original game. The pause system suspends the game and can be used whenever the player wants, allowing the player to use any item in their inventory and make split second decisions in real time.

This change alters the player's playstyle from strategic puzzle solving to a high-speed action playstyle.

Enemies

Guards are the primary obstacle, spawning in per-section groups called Units with predictable AI states that can be exploited. One guard patrols alone in a predetermined repetitive path, giving the player a window to strike without being seen. The core challenge loop is reading and exploiting these predictable patrol/detection patterns.

Playstyle

We want the player to either have a ghost or active playstyle, where the player will either try to sneak around the enemies or take out all the enemies before getting to the objective on the map.

The ghost playstyle is a playstyle of not wanting to be seen and not wanting to harm any enemies to achieve your objective. This fits the stealth genre of the game.

The active playstyle is a playstyle of where the player goes straight into the enemies and attacks them with their weapons and gadgets.

Items

The items that the player will have access to is limited as too many items will allow the player to play the game any way they want. By limiting the number of items to a specific number of unique items, the player is forced to play according to how we as the designers want them to play, which would be either the active or ghost playstyle.

To help enforce these playstyles the player only has the following weapons and gadgets:

- Longsword, Shortsword
 - These weapons were chosen to allow the player to easily kill enemies in their immediate range.
- Gun
 - To allow the player to kill enemies from long range.
- Concussion Hammer
 - Same as the longsword except that it doesn't kill the target allowing for the player to do capture missions.
- Slipstream
 - Allows the player to slow down time in order to easily get rid of groups of enemies.
- Teleporters
 - Different gadgets that take the player from one point to another.

What is being left out of this cloned system

Mission board

The player will not be able to choose their missions and will be forced to do missions that exist in the single large map we made. This will give the player a more hands-on experience as they will not know what to expect until they are given the mission, thus

making it the players job to find out what exactly they are supposed to do during the mission.

Ships

The player will not be able to fly towards the map, which removes the entire flying mechanic along with trying to get past the big ships' sensors. Will remove windows as an extra exit.

Shop

The clone will have no shop as there is no currency gained from mission completion. Therefore, players can only get items from chests on the map.

Other Items

Additional items were not added to make the hypothesis easier to test. This was also why the pause system was removed.

Game Research

Category and Sub-Category Analysis

To understand the process of cloning the game Heat Signature (2017), it is important to clearly define what type of game is being developed. Ernest Adams defines a genre as a category characterised by "particular kinds of challenge, regardless of setting or game-world content". Following this definition, this cloned game can broadly be categorised as a **stealth/shooter game**, as the player must strategically manoeuvre around the map to avoid or engage enemies.

However, genre classifications are often too general to be useful on their own, so it is necessary to break them down further into sub-genres, which define what the moment-to-moment gameplay of a game will be like (Vucetic, 2026). Within the broader shooter category, Heat Signature sits specifically within **2D Top-Down Shooter Games**, as both the player and enemies engage each other from a top-down perspective.

What is the game in these genres

There are several features that commonly define the 2D Shooter sub-genre that exist in Heat Signature and were replicated in our clone game, being an emphasis on facing overwhelming numbers of enemies, having different weapons suited to different situations (melee vs ranged weapons) and item collection/management. However, unlike other common 2D top-down shooter games, Heat Signature deliberately tracks limited item usage and cooldowns, and enemies are encountered in small, deliberately placed guard groups called 'Units' rather than overwhelming numbers. This separation

from other games signals that Heat Signature is not a pure 2D shoot-'em-up, but a hybrid that blends 2D shooter conventions with stealth-action genre challenges.

This stealth hybridity introduces further sub-genre-defining challenges, such as a vision-cone/FOV system, where the player must avoid the enemy's eyeline and navigate blind spots. The cover and spatial navigation, where obstacles like walls and doors block sightlines for both the player and enemies. Additionally, the stakes in Heat Signature are notably higher than in the average 2D top-down shooter, as a single hit dealt by either the player or an enemy result in instant permadeath of the character. This design choice is shared with games such as Hades and Enter the Gungeon, where high individual stakes exist to preserve the core challenge of the game without punishing the player's overall progress.

Therefore, Heat Signature and our clone game is a hybrid genre that blends 2D top-down shooter conventions (weapon variety, item management, enemy engagement) with stealth-genre challenges (vision-cone detection, cover-based navigation) and high-stakes, permadeath-driven tension.

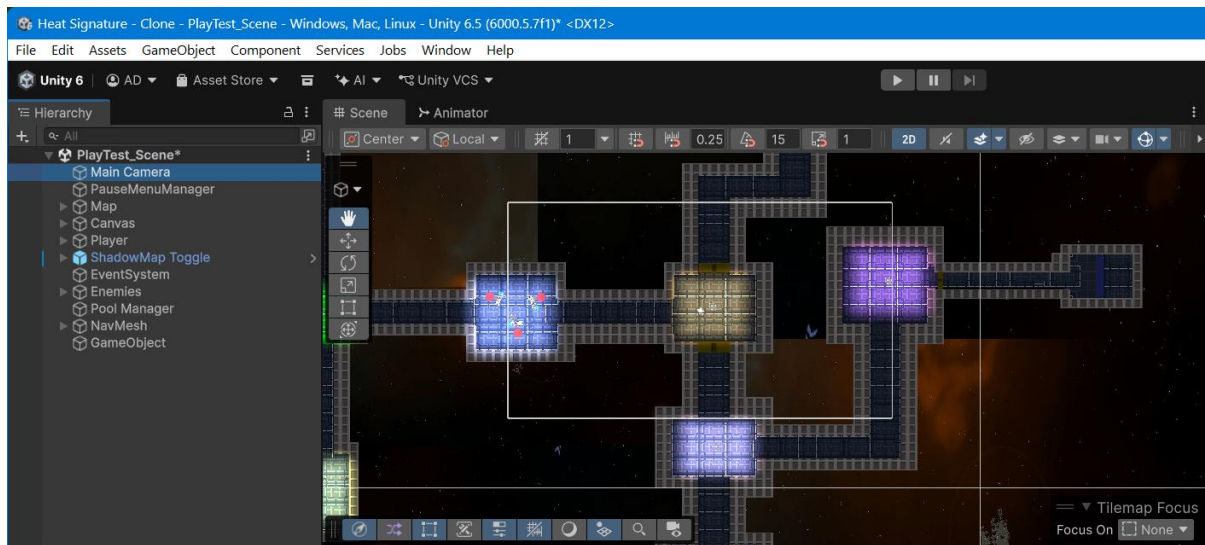
Iterative Process

The camera iterative process:

When the player and camera were initially implemented into the game, a 2D character controller with animations was included, and the camera followed the player at a 50-degree angle with a height of 12 above the player. We did this to recreate the movement and camera from Heat Signature. However, this CameraFollow script and character controller ended up looking more like the player and camera from 2D top-down games such as Legend of Zelda, rather than Heat Signature. Additionally, mechanics like the slipstream and teleport did not suit the current player character, making it difficult to implement these systems correctly.

As a result, the group decided to redo both the player and camera. The character was replaced with a simple triangle due to time constraints, allowing the movement mechanics to be implemented correctly without being tied to animation constraints. This was done to keep the game as close to Heat Signature as possible, so that our

hypothesis could be tested accurately.



(Figure 4; screenshot of the camera and the hierarchy window)

At first all the weapons could only be used by the player, as the weapons would search for the mouse position of the player and then find the target of the weapon actions. This was found as a bug when we added that to the enemy AI as everytime an enemy used a weapon, it would be inaccurate when they attempted to shoot the player.

This caused the code of the weapons to change to use input from the incorrect script to get the target of the enemy instead of the weapon script itself.

The original Time Manager script used to change the time of the game resulted in everything working at a slower time. This was changed to a unique variable on every object that uses the Time Manager to allow the player to still move at a normal time while everything else is slowed down.

Item pick up was removed from the clone, which in turn caused other mechanics to be removed due to them relying on item pick up. The list includes:

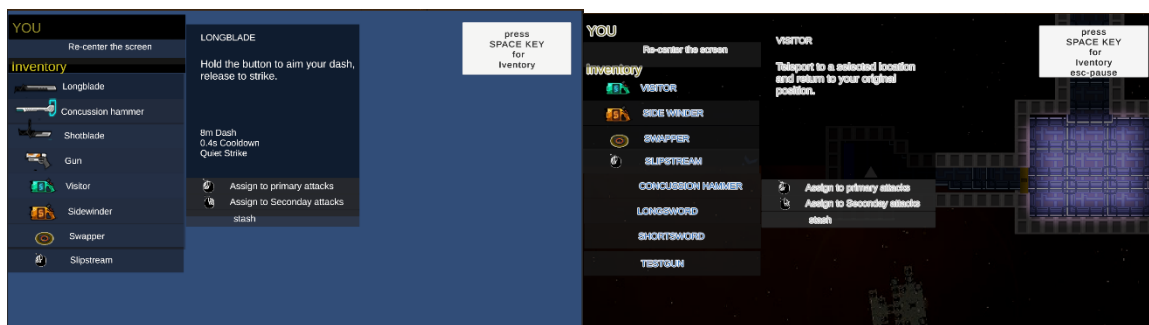
- Throwing Items.
- Chests that contain items
- Dropping Items.

The inventory iterative process;

The iterative process focused on developing the inventory system for the game, drawing inspiration from the inventory design of Heat Signature. The process began with creating the inventory UI with eight slots and implementing the item information panels. Item sprites and information were also added so that each gadget and weapon could be displayed correctly.

During testing, several issues were identified with how items were interacted within the inventory slots and mouse controls. The development process focused on making the first two slots usable with left and right click, while also allowing items to be moved between the inventory slots. These features were tested repeatedly, and the scripts were adjusted when certain items displayed incorrect information or did not respond correctly to mouse clicks.

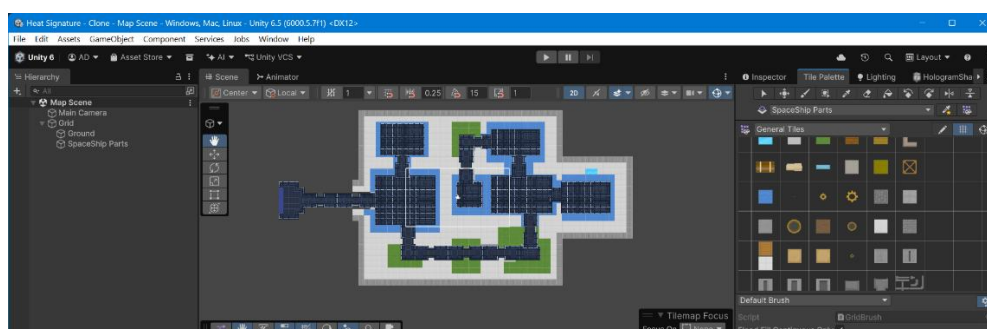
Through repeated testing, problem-solving, and adjustments, the inventory system was gradually improved. This process helped develop a better understanding of how the UI, inventory system, and item scripts needed to work together. The final inventory was therefore not completed in a single attempt, but developed through continuous testing, identification of issues, and implementation of solutions. Even though there were issues with the sprites of the weapons at the final creation of the inventory, we've decided that players will not have sprites intact for the weapons, only the the gadgets will have it intact.



(Figure 5; screenshot of the testing inventory and the final inventory)

Design evolution

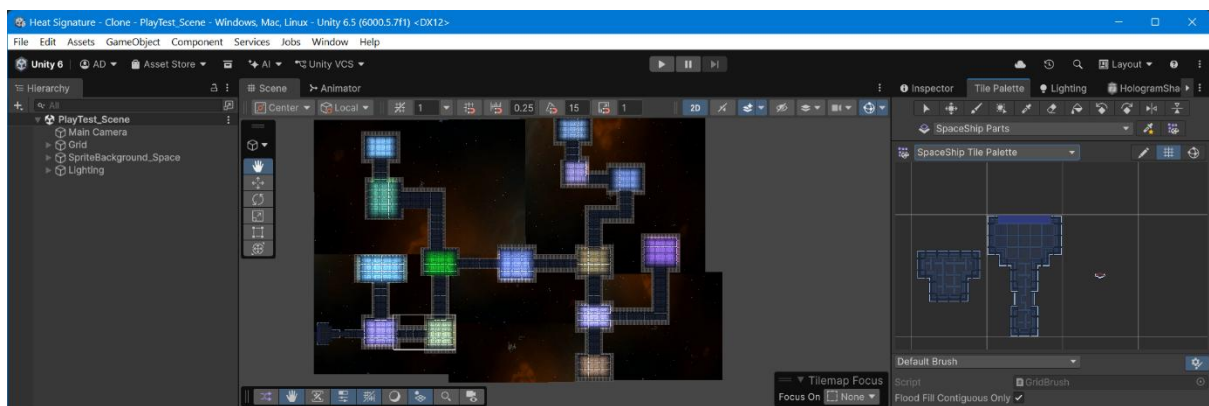
The design evolution of the clone game shows how our ideas developed from the initial concept to the final design. The process included an initial sketch of how the inventory would look, followed by the first map design and the tilemap used to create the environment.



(Figure 6; screenshot of the initial map design)

The first map design was mainly created to help the team become comfortable with the tools and understand how to build the game environment. At this stage, the team focused on developing their own design rather than directly referencing the visual style of Heat Signature. As the project progressed, they became more familiar with the game and began adapting their design to better reflect its visual style and atmosphere.

For the final map, one of our group members added a galaxy sprite and lighting to create a darker space-like atmosphere that resembles the feeling of Heat Signature. They also ensured that the map had a clear and organised layout, with each unit being easy to understand rather than placed randomly. This made the environment feel more intentional while still allowing the team to create their own interpretation of the original game.



(Figure 7; screenshot of the final map design, in playtest scene)

Conclusion

Overall, our group set out to clone the core mechanics of Heat Signature, with a specific focus on testing whether removing the pause-to-plan system and replacing it with the ShadowMap mechanic would shift the player's experience from tactical planning to in-the-moment decision-making. Throughout development, we prioritised keeping the clone as close to the original game as possible so that this hypothesis could be tested accurately. This shaped many of our design decisions, including limiting the player's available items and cutting systems such as the mission board, ships, and shop that were not essential to answering our core question.

The iterative process highlighted how an initial scope of a project will not always match the end result. The first camera and character controller ended up resembling a 2D top-down game like Legend of Zelda rather than Heat Signature, and several systems, such as the weapon-targeting logic and the Time Manager, required fixing once we discovered

that they did not function correctly for enemy AI or they did not produce the intended slowdowns across the whole game. These changes reflect our group's iterative process.

Through the category and sub-category analysis, we were also able to better understand Heat Signature not as a pure 2D shooter, but as a hybrid that blends shooter conventions with stealth-genre and action-genre challenges , which reinforced why removing the pause system specifically was a meaningful test of the game's core tension. Moving forward, our clone game gives us a functional base to evaluate our hypothesis further, particularly in how players adapt their playstyle, whether ghost or active, once they can no longer pause to plan their next move.

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